Mobile Application Development – CA1: Review 1

Idea & Proposal Review

Milestone: Problem definition, concept, and project scope

Team Members	1. Sonu Kumar Jha – 25MCAR0122 2. Abhinav Kumar Jha – 25MCAR0085
Title	Quiz Master – Smart Quiz and Assessment Management System
App Name	Quiz Master
Problem Statement	Traditional quiz systems often require manual question preparation, evaluation, and result calculation. Students may also face difficulty participating in quizzes simultaneously in a classroom or group environment. There is a need for a smart quiz application that provides interactive quizzes, automatic evaluation, time-based questions, instant results, and supports multiple students participating at the same time.
Abstract	Quiz Master is a mobile-based quiz and assessment application designed to make learning and testing more interactive, efficient, and accessible. The application allows users to select quiz categories, attempt multiple-choice questions, answer within a specified time, and receive instant results. It provides features such as countdown timer, automatic score calculation, correct-answer feedback, sound and vibration notifications, and category-based quizzes. The application can also support group quizzes with 70+ students simultaneously. An advanced feature allows users to upload Word, PPT, or document files and generate objective questions automatically based on the uploaded content, with options to select the number and difficulty level of questions.
Target Users	Students, teachers, educational institutions, trainers, and users preparing for academic and competitive examinations.
Planned Core Features	1. Category-Based Quiz 2. Multiple-Choice Questions with 4 Options 3. Countdown Timer 4. Automatic Score Calculation & Instant Results 5. Correct Answer Feedback 6. Sound & Vibration Notifications 7. Group Quiz with 70+ Students 8. Selection of Question Count & Difficulty Level 9. Word/PPT/Document Upload for Automatic MCQ Generation 10. Leaderboard & Performance Tracking
App Icon	Use a Quiz Master logo/icon featuring a question mark, quiz symbol, trophy, or smart-learning element.

Signature of Members	**Signature of Faculty In-charge**
Sonu Kumar Jha 25MCAR0122 Abhinav Kumar Jha 25MCAR0085	Mr. Aldrin Jose Antony Signature: 28/8/26

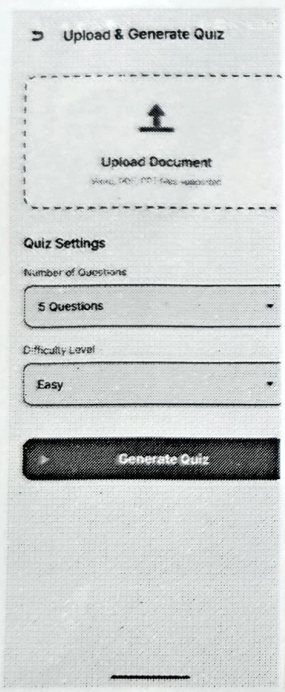
Mobile Application Development – CA1: Review 2
UI/UX Design Review

Milestone: Wireframes, mockups, and visual design language

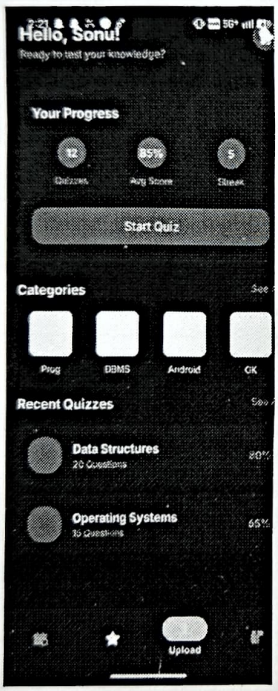
Team Members	*1. Sonu Kumar Jha – 25MCAR0122* *2. Abhinav Kumar Jha – 25MCAR0085*
Title	*Quiz Master - Smart Quiz and Assessment Management System*
App Name	*Smart Quiz Generator*
Aim	*To develop an advanced Android-based Quiz Application that can generate objective questions automatically from uploaded study materials such as PDF and Word documents.*
Objectives	*To provide a user-friendly online quiz platform.* *To allow users to upload PDF, Word and other supported study files.* *To extract useful content from uploaded documents.* *To automatically generate MCQ questions with four options from the uploaded content.* *To allow users to select the number of questions and difficulty level.* *To provide a countdown timer during the quiz.* *To calculate and display the final score.* *To maintain a leaderboard and user profile.*
Technologies Used	*Platform: Android Studio* *Language: Kotlin* *UI: XML / Android Views* *Database: Firebase* *AI/Question Generation: AI-based question generation* *IDE: Android Studio*
Main Features	*User Authentication – Login/Register functionality.* *Quiz Categories – Select different quiz subjects/categories.* *File Upload – Upload PDF, Word or supported study documents.* *Question Generation – Generate MCQs automatically from uploaded content.* *Quiz Configuration – Select number of questions and difficulty level.* *MCQ Quiz – Four options for each question.* *Timer – Countdown timer with time-up functionality.* *Result & Score – Automatically calculate the obtained score.* *Leaderboard – Compare scores and rankings.* *Profile – Display user information and performance.*
Working Flow	*Login → Home → Upload PDF/Word → Extract Content → Generate Questions → Select Questions/Difficulty → Start Quiz → Timer → Submit/Time Up → Result → Leaderboard*
Result	*The advanced Quiz Application successfully provides an interactive quiz environment where*

	users can upload study materials and generate objective questions automatically from the uploaded content. The application also supports quiz configuration, timer, scoring, leaderboard and profile features.
Conclusion	*The proposed application combines Android development and AI-based question generation to create a smart and flexible learning platform. It reduces the effort of manually creating quiz questions and allows students to generate quizzes directly from their study materials.*

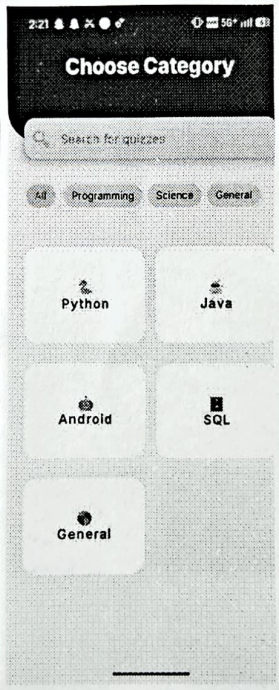
Sample UI Screenshots



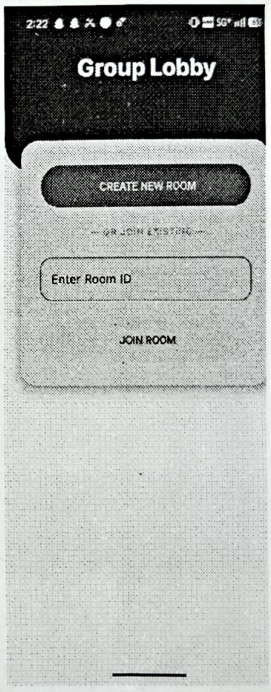
Upload & Generate Quiz



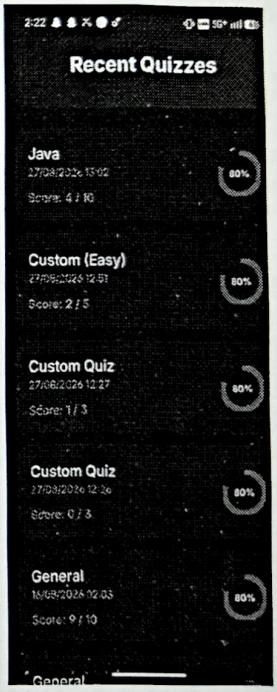
Home Dashboard



Choose Category



Group Lobby (Multiplayer)



Recent Quizzes / Results

Evaluation – UI Design & Visual Appeal

Remarks:

Sonu Kumar Jha (25MCAR0122)
Abhinav Kumar Jha (25MCAR0085)

Signature of Members

28/8/26

Signature of Faculty In charge
Mr. Aldrin Jose Antony

Mobile Application Development – CA1: Review 3

Development & Responsiveness Review

Milestone: Core build progress and responsive layout implementation

Team Members	1. Sonu Kumar Jha — 25MCAR0122 2. Abhinav Kumar Jha — 25MCAR0085
Title	Quiz Master – Smart Quiz and Assessment Management System
App Name	Quiz Master (Smart Quiz Generator)
Features Implemented	Login and Registration screens; Home Dashboard with progress overview; Upload & Generate Quiz screen (PDF/Word/PPT upload); Choose Category screen; Quiz screen with countdown timer and MCQ options; Result and score screen; Group Lobby for multiplayer quizzes; Recent Quizzes/Results history screen; Firebase integration structure; and responsive layouts for different Android screen sizes.
Tech Stack Used	Android Studio, Kotlin, XML/Android Views, Firebase Authentication, Firebase Firestore/Realtime Database, Python/Flask backend for AI-based question generation, NLP-based document parsing, REST API and Room Database.
Devices / Screen Sizes Tested	Android Studio emulator using compact and standard Android phone profiles. Small, medium and large phone layouts and portrait orientation are considered for responsiveness testing.
Responsive Design Approach	Constraint-based layouts, ScrollView for longer content, dp/sp dimensions, flexible cards, scalable text and consistent margins are used so the application adapts to different screen sizes without clipping or horizontal overflow.
Known Issues / Bugs	End-to-end AI question-generation backend integration, multiplayer group-quiz room synchronization and some edge/error states still require final verification. Minor UI spacing and responsiveness issues may be refined before the next review.
Add-ons as per Remarks	Improve loading/error states during document upload and question generation, refine leaderboard visualization, strengthen notification handling and add clearer quiz-generation status indicators based on faculty feedback.

Evaluation – Responsiveness & Interactivity:

Remarks:

17/9/26

Mr. Aldrin Jose Antony

Signature of Faculty In charge

Sonu Kumar Jha (25MCAR0122)
Abhinav Kumar Jha (25MCAR0085)
Signature of Members

Mobile Application Development – CA1: Review 4

UX, Navigation & Usability Testing

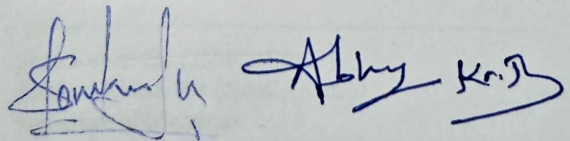
Milestone: User flow, interactivity, and usability feedback

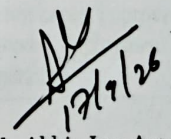
Team Members	1. Sonu Kumar Jha — 25MCAR0122 2. Abhinav Kumar Jha — 25MCAR0085
Title	Quiz Master – Smart Quiz and Assessment Management System
App Name	Quiz Master (Smart Quiz Generator)
Navigation Flow Summary	Splash Screen → Login/Register → Home Dashboard → Upload & Generate Quiz / Choose Category → Quiz Configuration (question count & difficulty) → Quiz (Timer) → Result & Score → Leaderboard. Group Lobby allows creating or joining a room for multiplayer quizzes. Back navigation returns to the previous logical screen.
Interactive Components Used	Buttons, text fields, password fields, document upload control, category cards, dropdown selectors (question count/difficulty), countdown timer, MCQ option buttons, room-ID input, create/join room buttons, leaderboard list, progress indicators and scrollable content.
Usability Testing Method	Peer testing with the two team members and selected student users. Testers perform common tasks such as login, uploading a document to generate a quiz, selecting category/difficulty, attempting a timed quiz, creating/joining a group quiz room and viewing the leaderboard.
Usability Feedback Received	The interface should keep the upload-to-quiz flow simple, make category icons and difficulty selection clearer, keep the timer prominent, improve readability of generated questions and scores, and avoid overcrowding the home dashboard.
Navigation Improvements Planned	Reduce unnecessary transitions between upload and quiz start, keep important actions within one or two taps, improve room joining and error handling, retain bottom-navigation state, and simplify the result and leaderboard screens.
Add-ons as per Remarks	Add onboarding/help text for document upload, accessibility labels, confirmation dialogs where needed, clearer notification and error messages, and a visible disclaimer noting that AI-generated questions may require review for accuracy.

Evaluation – User Experience & Navigation

Remarks:

Sonu Kumar Jha (25MCAR0122)
Abhinav Kumar Jha (25MCAR0085)
Signature of Members





Mr. Aldrin Jose Antony

Signature of Faculty In charge

Mobile Application Development – CA1: Review 5

Final Presentation & Deployment Review

Milestone: Completed app, deployment, and final demo

Team Members	1. Sonu Kumar Jha — 25MCAR0122 2. Abhinav Kumar Jha — 25MCAR0085
Title	Quiz Master – Smart Quiz and Assessment Management System
App Name	Quiz Master (Smart Quiz Generator)
Deployment Details	Final Android APK will be generated from Android Studio for demonstration and testing. The AI/NLP-based question-generation backend will be hosted on a suitable Python server, with Firebase used for authentication, cloud data storage and real-time synchronization. Final APK/repository/hosting links will be added before submission.
Final Feature Summary	User authentication; category-based quizzes; document upload (PDF/Word/PPT) with automatic MCQ generation; configurable question count and difficulty level; countdown timer; automatic score calculation and instant results; correct-answer feedback; sound and vibration notifications; group quiz supporting 70+ students; leaderboard and performance tracking; user profile; responsive UI.
Demo Notes	Live demo flow: Login → Home Dashboard → Upload study document → AI-based question generation → select question count/difficulty → Start Quiz → Timer → Submit → Result/Score → Leaderboard. Multiplayer flow: Create/Join Group Room → simultaneous quiz with multiple students → shared results. Responsive layout will also be demonstrated.
Project Report Submitted	Yes — final report will include project overview, problem statement, objectives, architecture, UI/UX, AI/question-generation workflow, database design, implementation, testing, screenshots, limitations and future enhancements. Final links will be attached according to faculty instructions.
App Icon	
Add-ons as per Remarks	Incorporate final faculty-requested UI/UX changes, additional test cases, improved error handling, final evaluation of the AI question-generation model, deployment verification, documentation updates and final presentation preparation.

Final Evaluation – CA1 Rubric Total 15 Marks

Remarks:

12/9/26

Mr. Aldrin Jose Antony

Signature of Faculty In charge

Sonu Kumar Jha (25MCAR0122)
Abhinav Kumar Jha (25MCAR0085)
Signature of Members